Newtonian Mechanics And Gravitation

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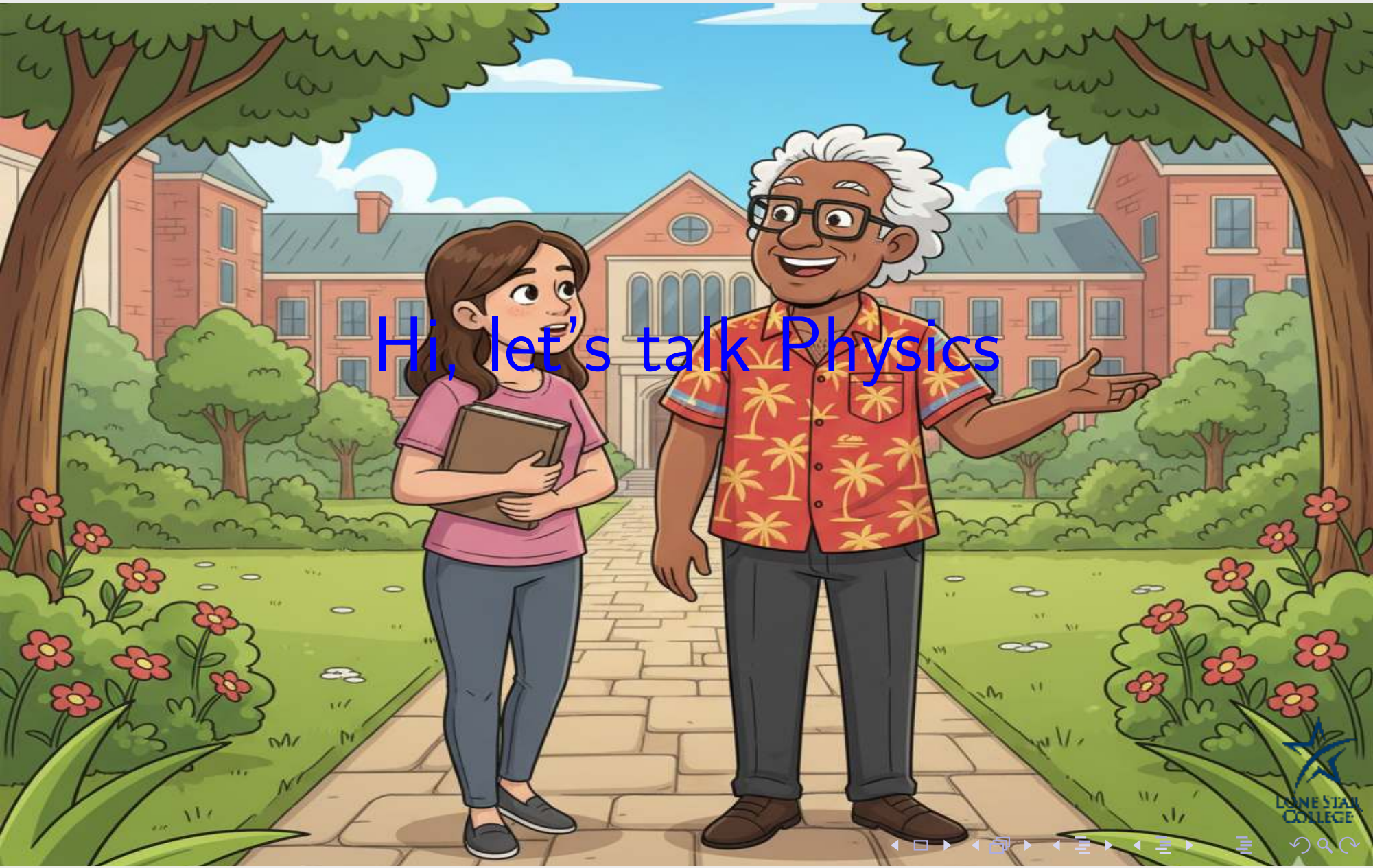

Hi, let's talk Physics

What is This Thing Called Dynamics?

- So far, we have focused on **Kinematics**: the mathematical description of motion (position, velocity, and acceleration) without considering what causes it.
- We can describe *how* an object moves, but we don't yet have the slightest idea about *why* it moves.

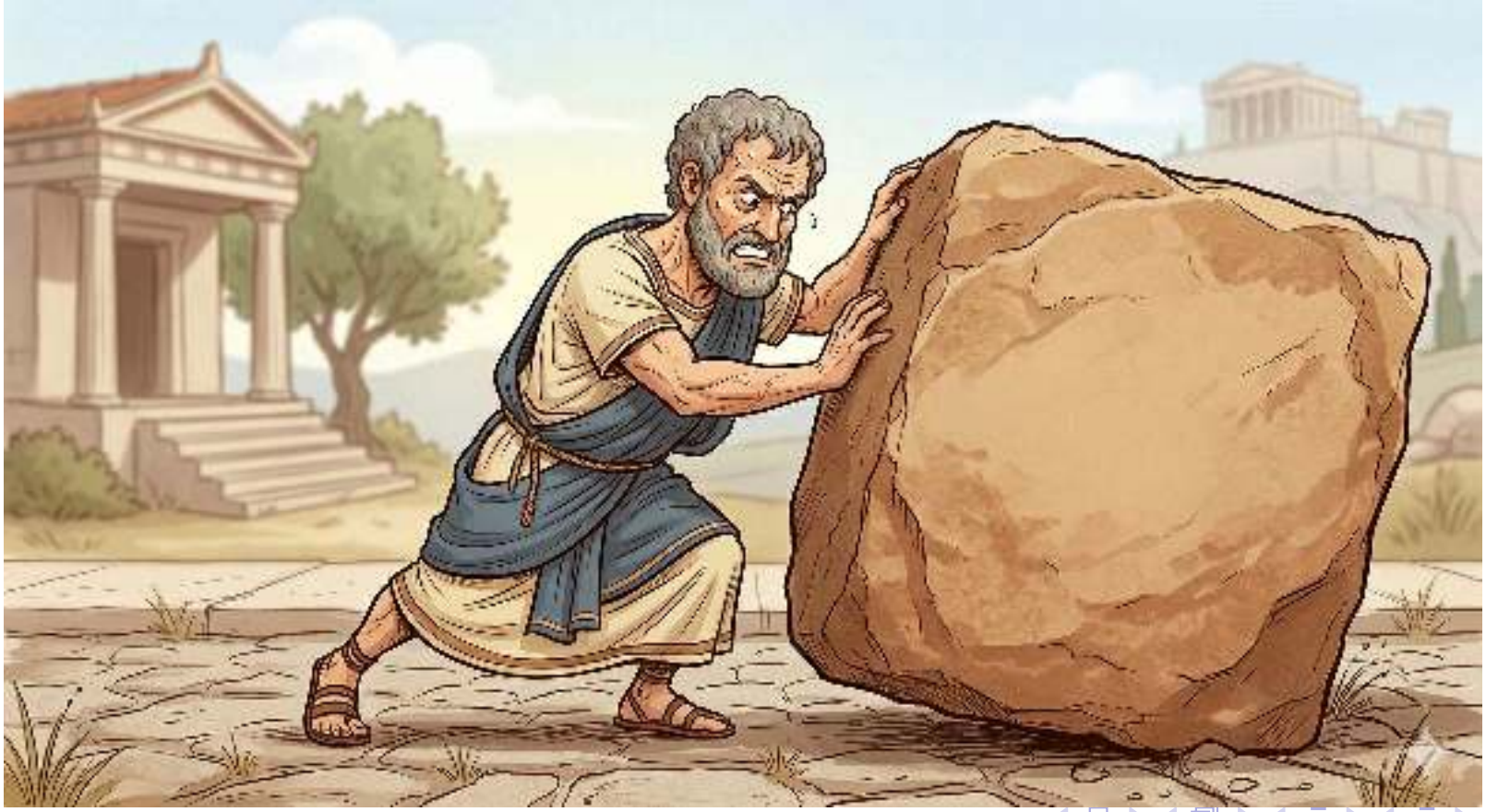
Dynamics

Dynamics is the branch of mechanics that investigates the **causes** of motion. It explains how interactions between objects—which we call **forces**—change their state of motion.

"In Kinematics, we are spectators; in Dynamics, we find the protagonists."

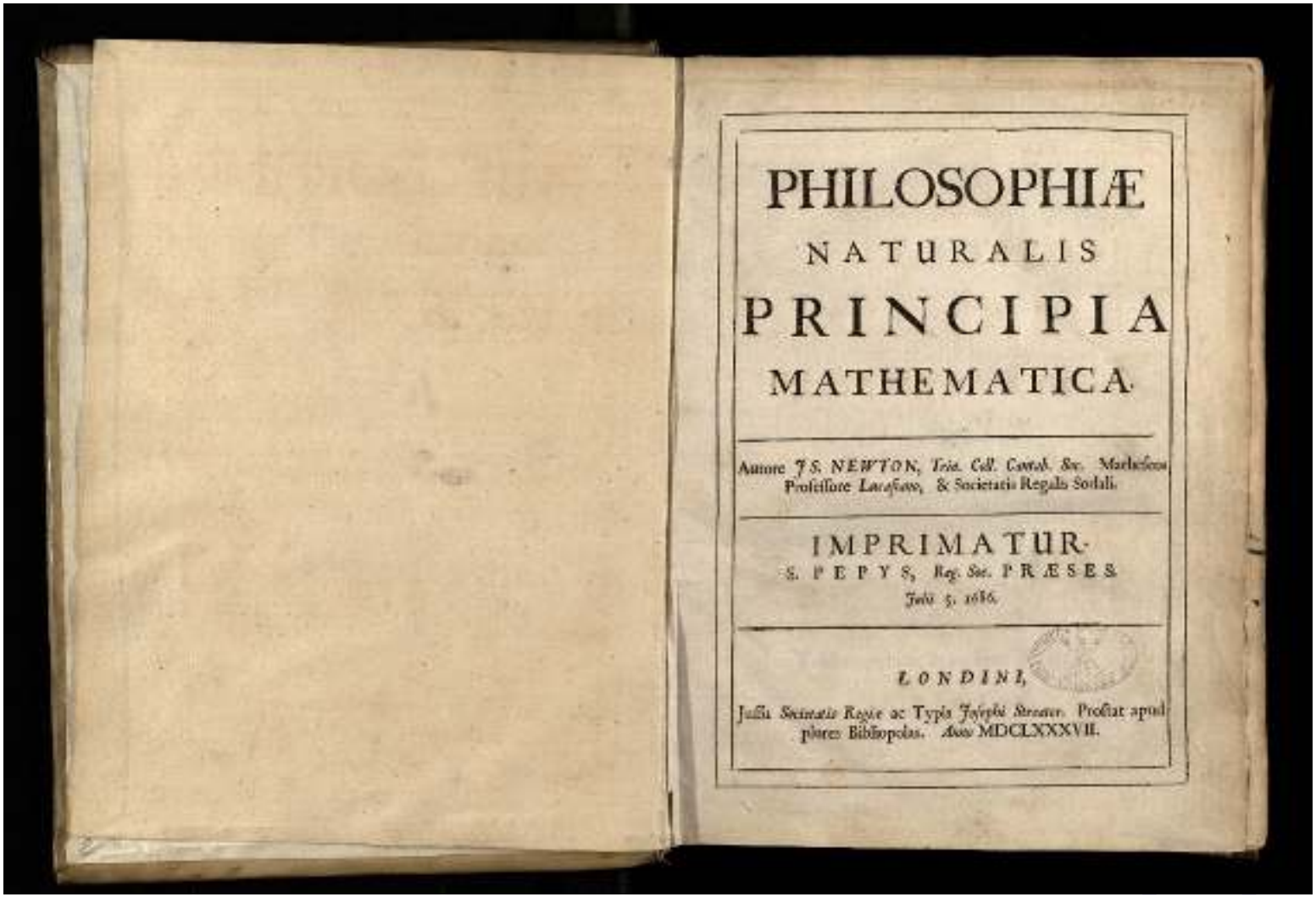
How is motion produced?

Aristotle, who by the way was Alexander The Great mentor, thought that objects need to be pushed to sustain their motion



Aristotle's ideas prevailed from around 323 BC to 1613 AD when Galileo Galilei revolutionized physics by establishing that motion is relative, objects in motion stay in motion (inertia), and all objects fall at the same rate regardless of mass.

Mechanics got its curent form when in 1686 Isaac Newton published one of the most important writings in history:



PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore JS. NEWTON, Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.

IMPRIMATUR.
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.

LONDINI,
Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures Bibliopolas. Anno MDCLXXXVII.

Newton's Laws

Established in the Principia, they are

- **Law of Inertia** An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
- **Law of Acceleration (or Force)** The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass ($\vec{F}_{net} = m\vec{a}$).
- **Action-Reaction Law** For every action, there is an equal and opposite reaction. If Object A exerts a force on Object B, Object B simultaneously exerts an equal force in the opposite direction on Object A.

Thank to Zeus I was Wrong!

Why?

Why do we need to wear the seat belt

Answer: Inertia (First Law)

Contact and Reaction Forces

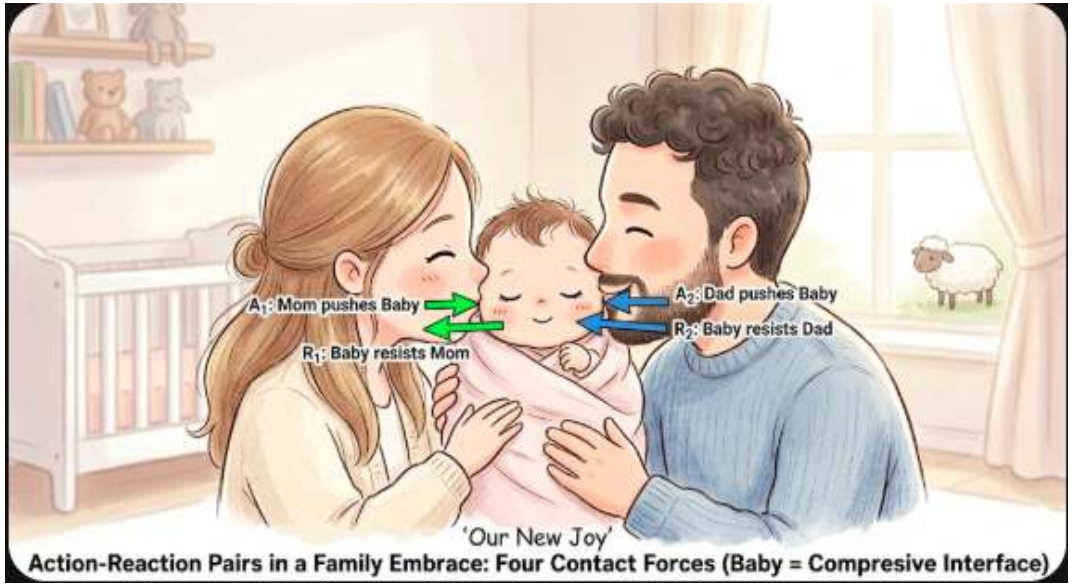
The wall exerts a force on the head of the soldier, it is the reaction of the force that the head does on the wall

Reaction on the soldier $\vec{\Lambda} = \vec{\lambda}_{||} + \vec{\lambda}_{\perp}$

Contact Forces: Beware of Multiple Agents

Describe the contact forces associated with the kisses

Contact Forces Beware of Multiple Agents



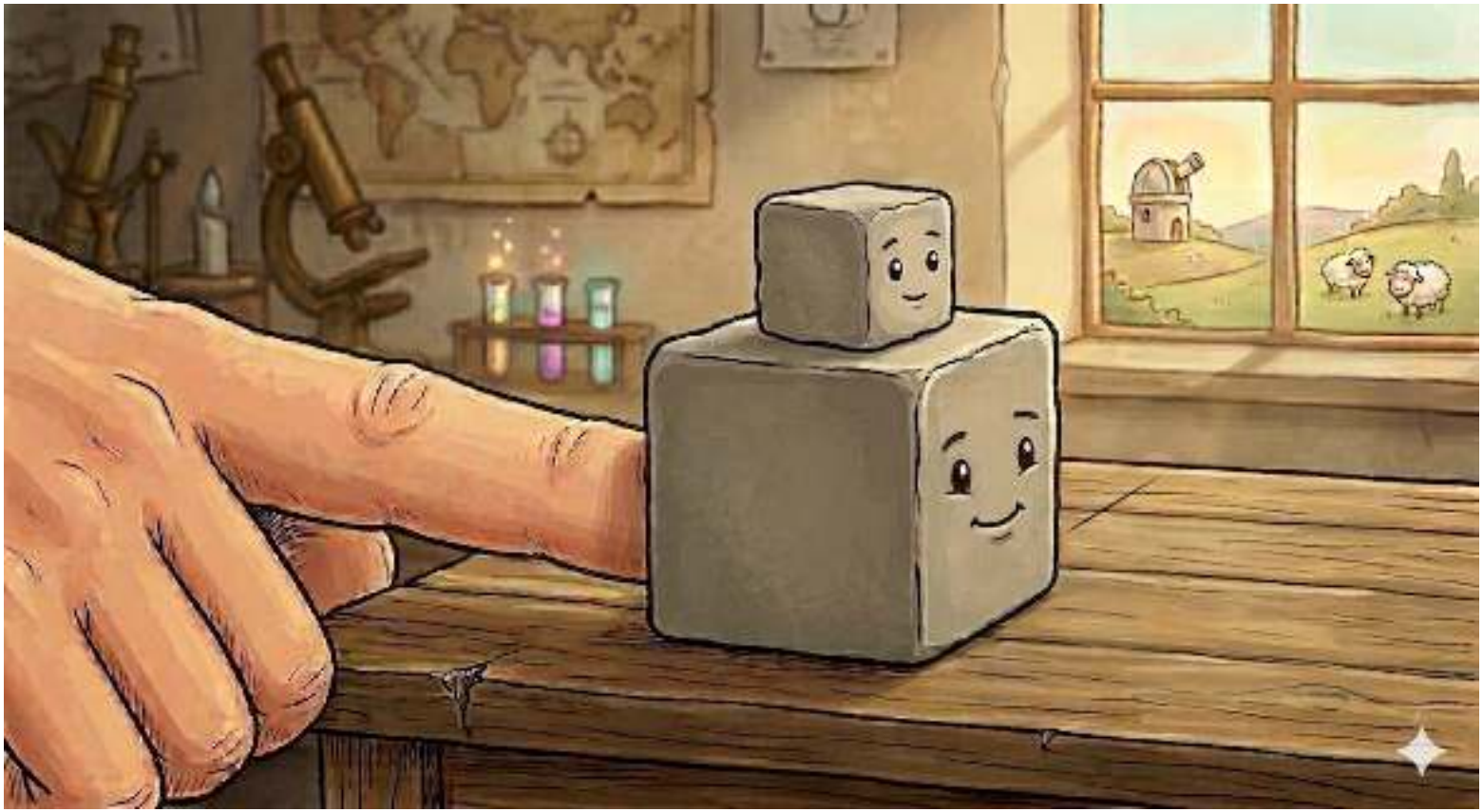
Action-Reaction Pairs in a Family Embrace: Four Contact Forces (Baby = Compresive Interface)

Reactions

$$\vec{\Lambda}_{Mom,Baby} \Longleftrightarrow \vec{\Lambda}_{Baby,Mom}$$

$$\vec{\Lambda}_{Dad,Baby} \Longleftrightarrow \vec{\Lambda}_{Baby,Dad}$$

The finger pushes with force $\vec{F}$ and the two little blocks move together, there is no friction with the table. (1) Explain the motion of the little block and (2) What is its acceleration?



Let the mases of the two little blocks be M_B and M_S (for bigand small) they together behave as one unit so the acceleration of the comoving masses satisfies

$$\vec{F} = (M_B + M_S)\,\vec{a} \rightarrow \vec{a} = \frac{\vec{F}}{M_B + M_S}$$

$\vec{a}$ is -clearly- the acceleration of the little block.
Now, there must be a horizontal force acting on the little block and it is not the push of the finger (the finger is pushing the big block), the only possibility is that there is friction between the two blocks and the friction of the big block on the small one is the agent accelerating the small block

Deepening Our Understanding I

Let's try to make sure we get everything. What forces act on the big block?

Answer

1. $\vec{F}$
2. The weight of the big block which is the gravitational force the Earth excerts on M_B
3. The force that the table exerts on M_B. It must be there or nothing would compensate the weight. This is called the reaction force $\vec{\Lambda}_{Table,M_B}$
4. The contact (reaction force) $\vec{\Lambda}_{M_S,M_B}$ that the litle box exerts on the big one.

Deepening Our Understanding II

1. $\vec{\Lambda}_{Table,M_B}$ being a vector may have horizontal and vertical components, i.e.

$$\vec{\Lambda}_{Table,M_B} = \lambda_{Table,M_B x}\hat{\mathbf{i}} + \lambda_{Table,M_B y}\hat{\mathbf{j}}$$

2. The same happens with

$$\vec{\Lambda}_{M_S,M_B} = \lambda_{M_S,M_B x}\hat{\mathbf{i}} + \lambda_{M_S,M_B y}\hat{\mathbf{j}}$$

3. According to Newton's third Law, there is another force, this one exerted by M_B on M_S given by

$$\vec{\Lambda}_{M_B,M_S} = -\vec{\Lambda}_{M_S,M_B}$$

Deepening Our Understanding III

The Newton's Third Law pair

$$\vec{\Lambda}_{M_B,M_S} = -\vec{\Lambda}_{M_S,M_B}$$

is responsible for the horizontal motion of the small mass. Indeed, on the small block, there are only two forces acting: its weight and the reaction force of the large block on it.
The vertical component of $\vec{\Lambda}_{M_B,M_S}$ compensates for its weight, while the horizontal component pushes the small block to the right. This is the agent causing its acceleration!
The horizontal component of the reaction is the friction between the two surfaces. Since they are not in motion with respect to each other, this agent is, in fact, **static friction**.

The General Strategy

How to attack any Newtonian Mechanics situation:

1. **Set the Stage:** Choose the observer and the reference frame.
2. **Isolate and Identify:** Look at each body and identify the set of all forces acting on it (Free Body Diagram).
3. **Apply the Law:** Write $\sum \vec{F} = m\vec{a}$ for each body.
4. **Identify Constraints:** Include any special conditions, such as an acceleration being zero in a specific direction.
5. **Solve:** Put all the equations together to form a system of equations and *let math do the rest!*

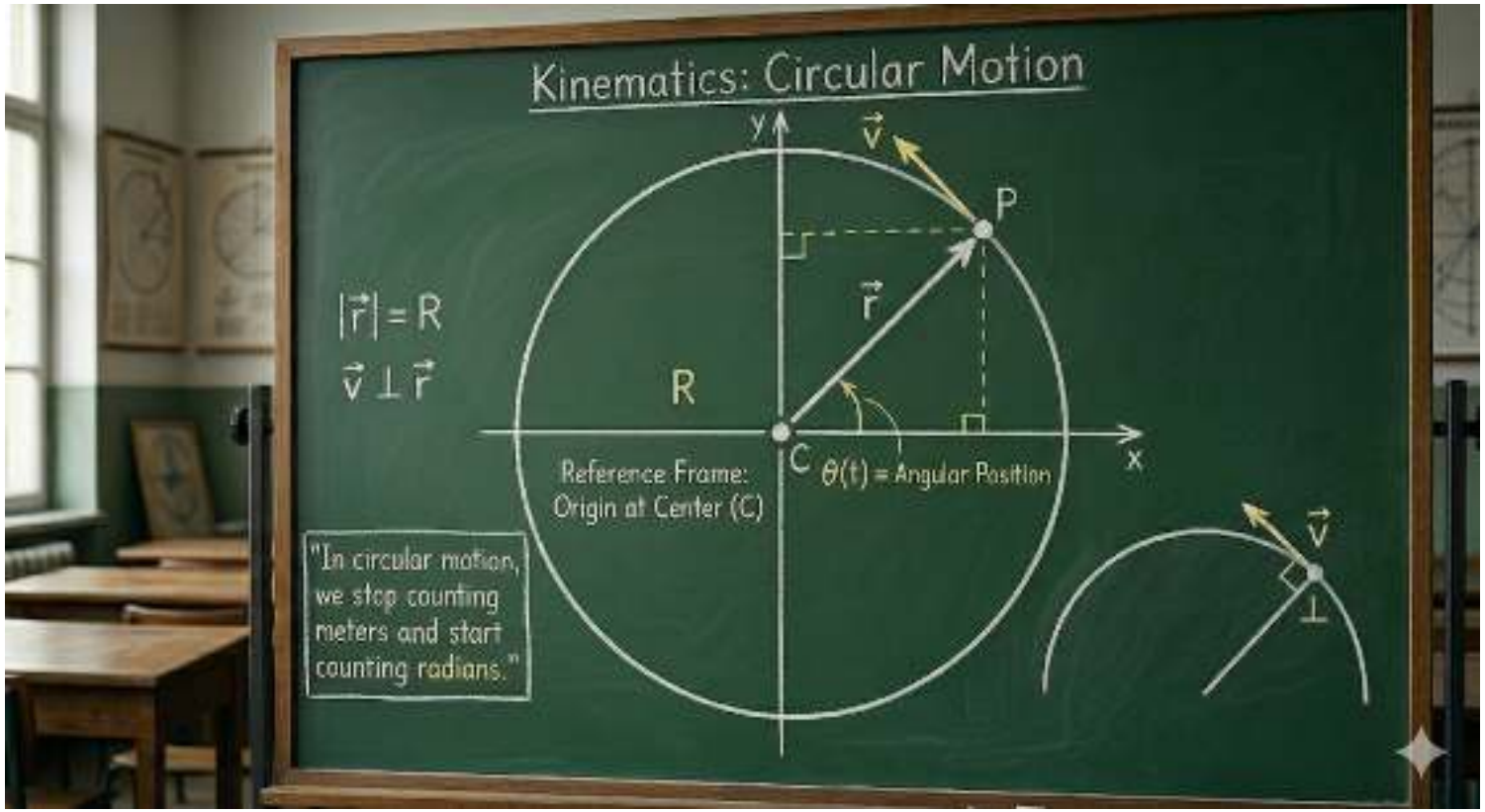
Kinematics:Circular Motion

Circular motion occurs when the **trajectory** of the moving object is a circle or a circular arc.

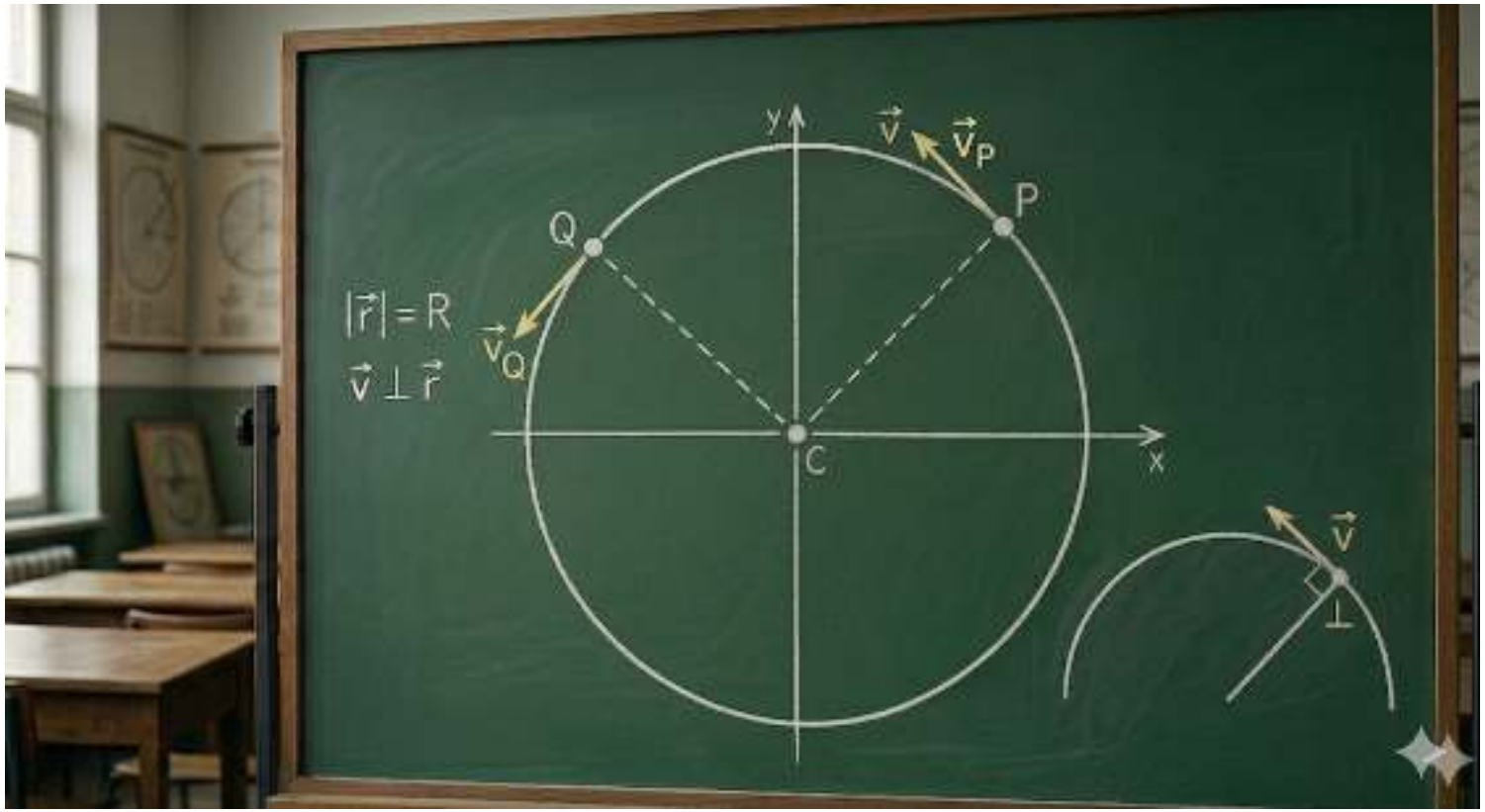
- **The Strategic Choice:** To simplify the description, it is convenient to place the origin of the reference frame at the **center of the circle**.
- **The Position Vector:** Under this choice, the position vector $\vec{r}$ always coincides with the radius of the circle ($|\vec{r}| = R$).
- **Angular Description:** Since the magnitude of $\vec{r}$ is constant, the motion is entirely described by the **angle** θ swept by the vector over time.

In circular motion, we stop counting meters and start counting radians.

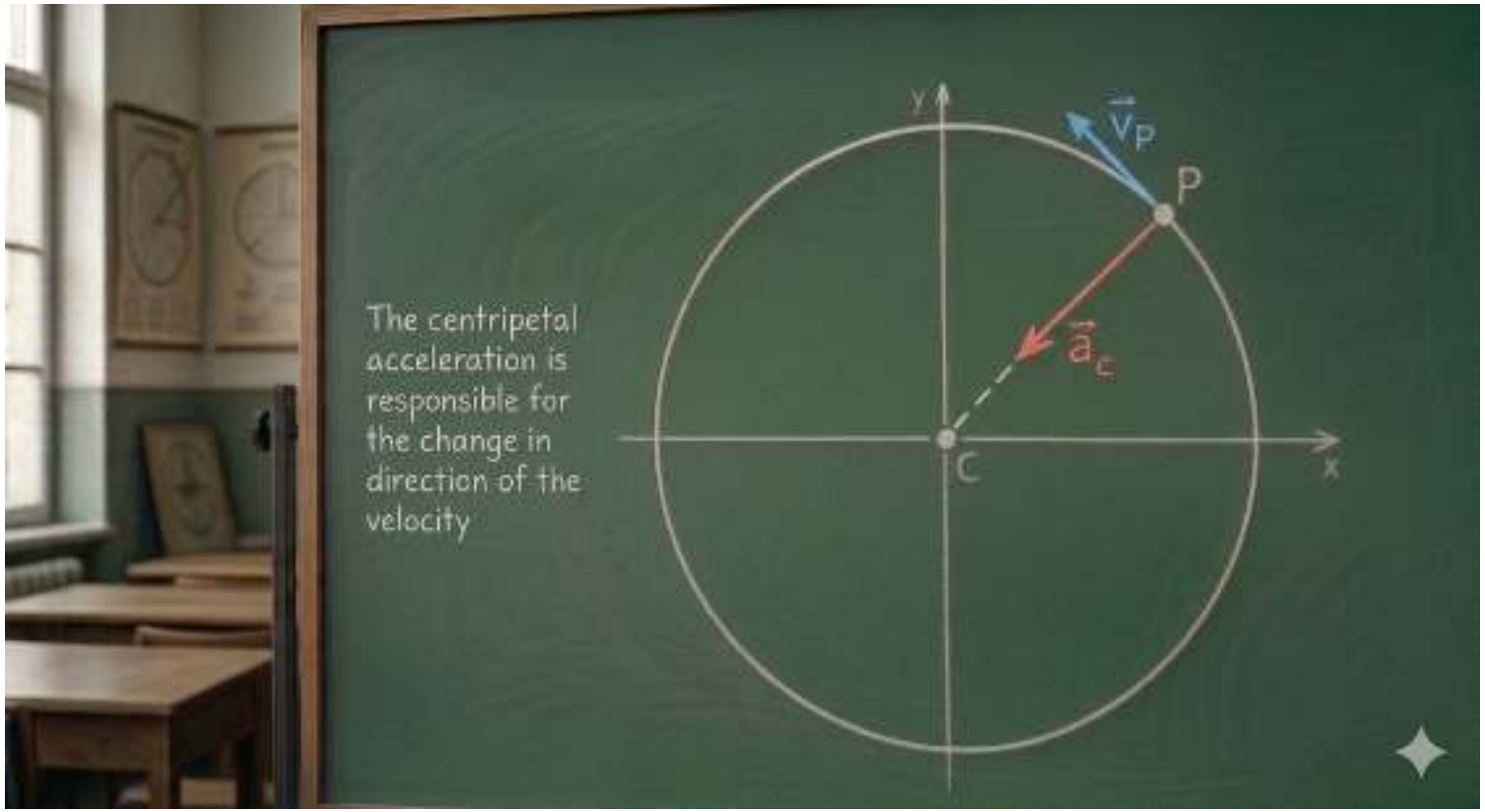
An obvious but fundamental observation, since the position vector is radial, and velocities are always tangent to trajectories, for circular motion, the velocity is always perpendicular to the position vector



The *tangential* velocity changes direction all the time so, even if the motion were uniform -in the sense that the speed were constant-, particles in circular motion are always being accelerated



At any point of the trajectory there is always acceleration that may be decomposed on a tangential and radial parts. The radial part -called centripetal acceleration $\vec{a}_c$- is always oriented towards the center



Example

Kinematics of a litle box on top of a table moving along a circular trajectory

To put math in action we need to learn that to measure angles in radians all that is needed is what follows.
Given an angle one draws an arch of circle subtending such angle and measure its radius r, then measure the arc length (ℓ), is the quotient

$$\theta = \frac{\ell}{r}$$

is the angle in radians, thus for example, if the angle is 360°, then the arc subtended would be a complete cirlce and since form midle school we now the circumference of a cirlce is

$$\mathcal{C} = 2\pi \times (\text{radius of the circle}),$$

then we will find the corresponding angle measured in radians is

$$\theta = 2\pi$$

A Vital Formula

Magnitude of the centripetal acceleration

$$\boxed{a_c = \frac{v^2}{r}}$$

where, of course, r is the radius of the circular trajectory

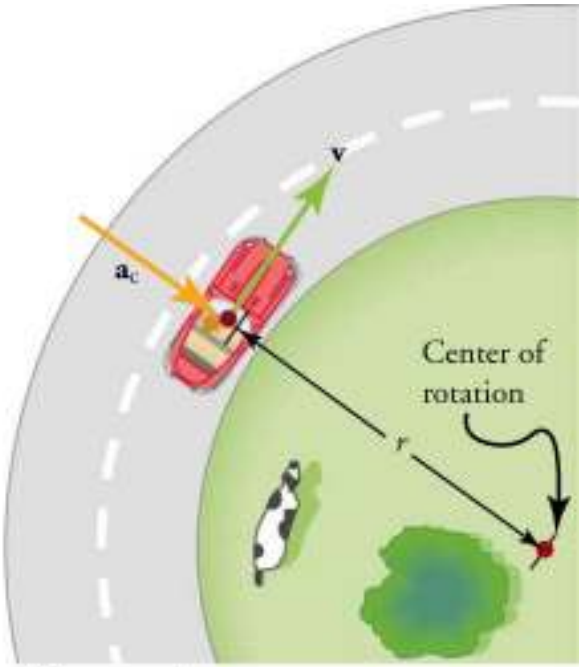
Drivers Remember

Note that a_c increases with v^2 and with $1/r$. That is why curves are almost always designed to be ample

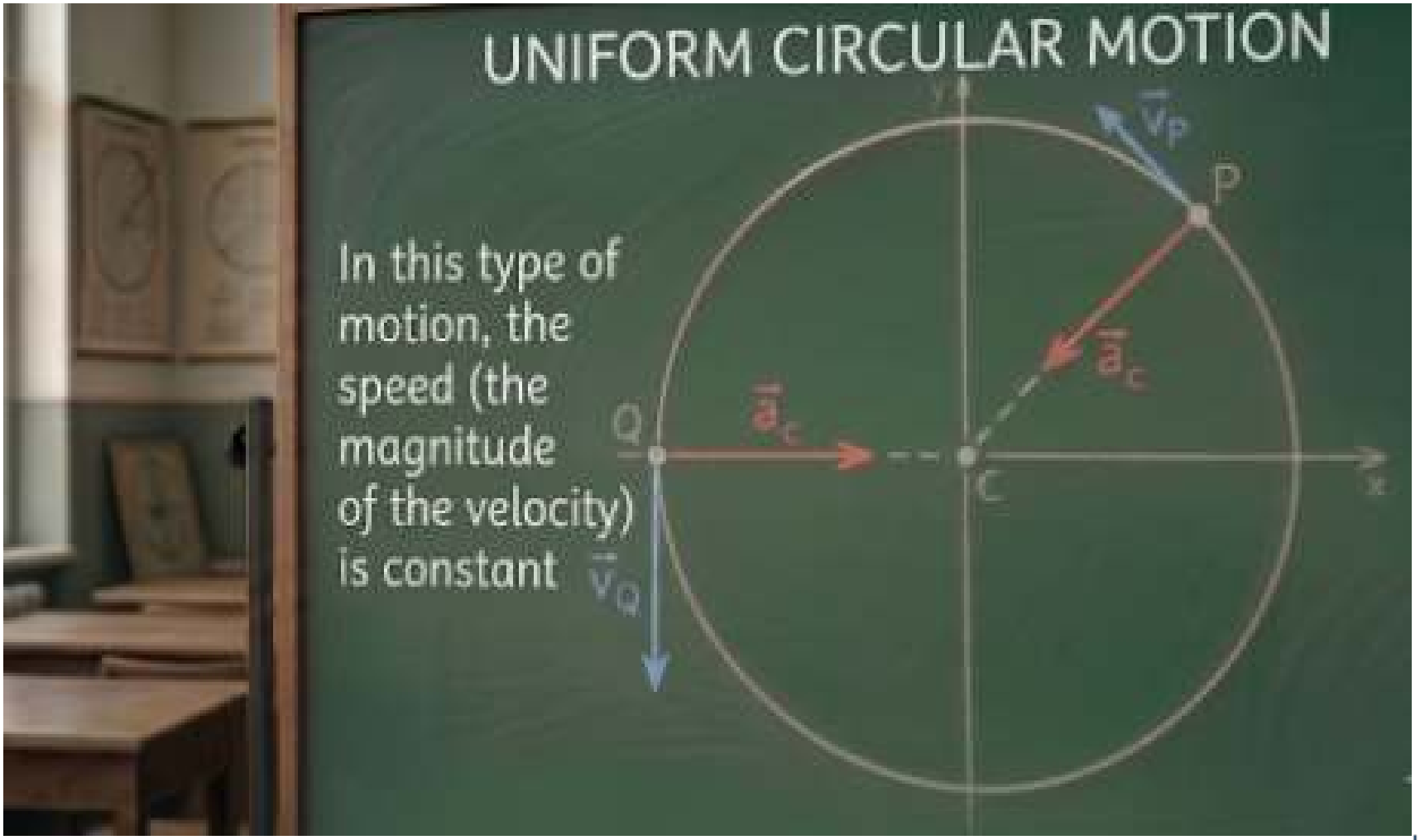
Beware

Drivers Remember

Never attempt to take a small radius curve -a corner for example- at high speed



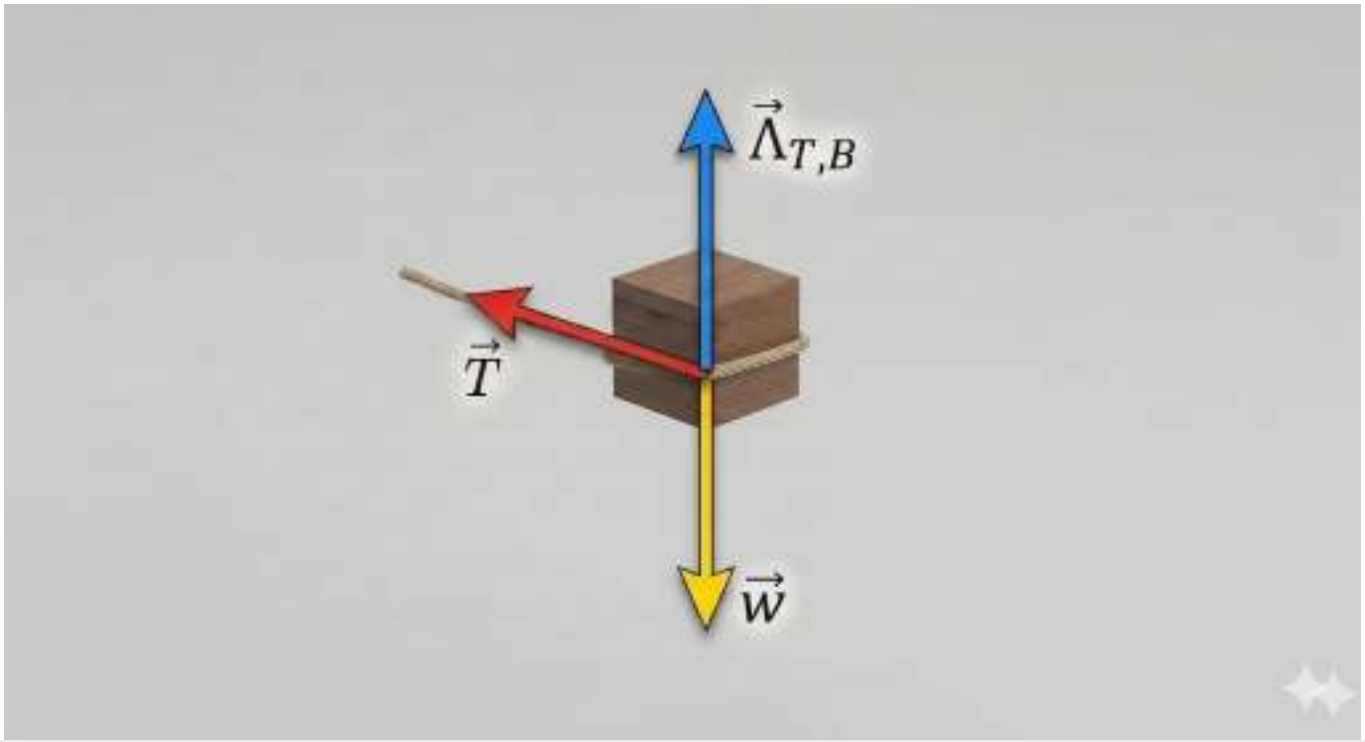
Car around corner


UNIFORM CIRCULAR MOTION
In this type of motion, the speed (the magnitude of the velocity) is constant
P
Q
C
x
y

Let's go back to our little box example, there is no friction between the table and the box and a string ties the box to the center of the table, the box moves at constant speed $|\vec{v}_t|$

We may isolate the little box on our minds to draw **all forces** acting on it, this is called a **free force diagram** the reaction of the table: its weight (the force exerted by the eath because of gravity) $\vec{W}$. $\vec{\Lambda}_{T,B}$ the reaction force of the table on the box, and $\vec{T}$ the tension exerted by the string, with these at hand we may write:

$$\boxed{\vec{W} + \vec{\Lambda}_{T,B} + \vec{T} = m_B\,\vec{a}_c}$$



Modelling the Situation

1. We have all the facts:
2.
   1. The speed $|\vec{v}_t|$ is constant so there is no tagential acceleration
   2. There is no friction so there is no component of the reaction $\vec{\Lambda}_{T,B}$ in the plane of the motion, or said differently, $\vec{\Lambda}_{T,B}$ is vertical
   3. The tension is parallel to the string, i.e. is radial
3. All the above summarizes in

$$|\vec{\Lambda}_{T,B}| = |\vec{W}| = m_B g$$

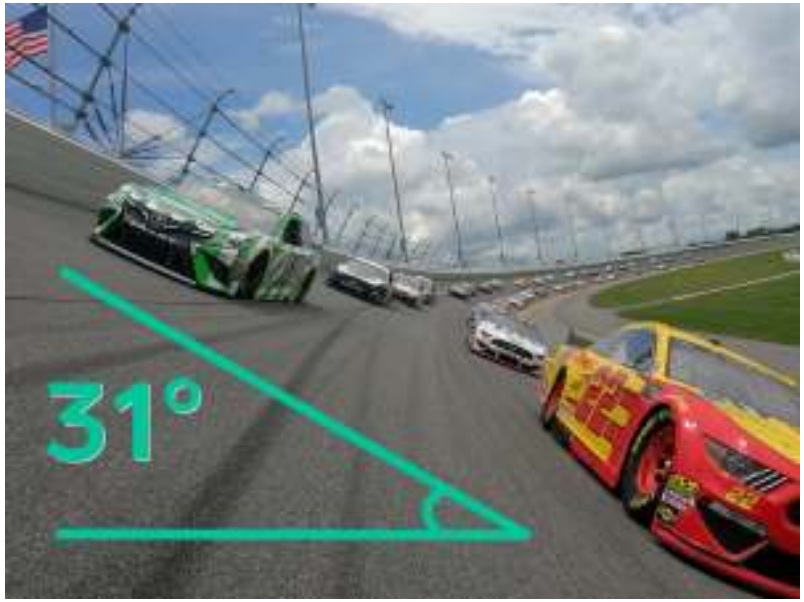
$$|\vec{T}| = m_B \frac{v_T^2}{r}$$

These two formuls arrived at by following a strict but simple logic, completely describe the system under study.

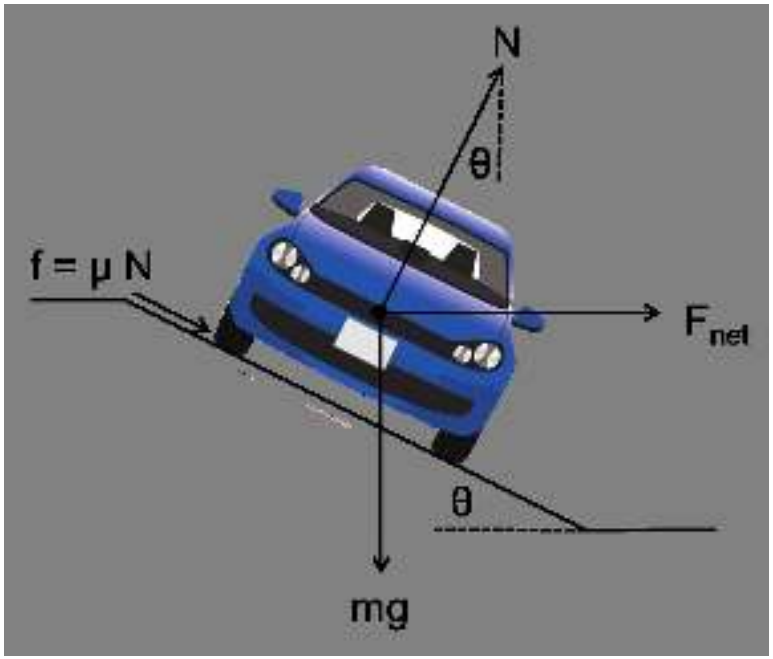
QUESTION

Why are NASCAR high speed curves banked

"The banking makes the reaction force of the track on the car to have a radial part that adds to the centripetal force"



Banking



Forces and their decomposition

Inertial Forces

We have all felt that, when riding a merry-go-round that is going fast, we seem to be violently pushed away from it. However, that is simply not true. What we are feeling is the tendency of our bodies to continue in a straight line as dictated by inertia.

But since we **doggedly** want to keep turning, an agent must exert a force on us to maintain the motion; i.e., some agent provides the force that sustains our centripetal acceleration. That sensation is called an **inertial** or **fictitious force** because such a force does not exist in an inertial frame. All we are feeling is the result of being observers while we ourselves are being accelerated.

Newton's Second Law and Inertial Frames

The Golden Rule

Newton's Second Law, $\vec{F} = m\vec{a}$, is only valid in **inertial reference frames** (non-accelerating systems).
In inertial frames, inertial or "fictitious" forces do not exist.

Enjoying Inertial Forces

- **The Reality:** When one rides *Formula Rossa*, the famous Abu Dhabi roller coaster, one gets submitted to an acceleration equal to 4.8 times g , the *real* force is the one pushing the car forward, and the seat pushing your back.
- **The Perception:** You feel a brutal "push" backward, but that is just your body's inertia resisting the change in motion.
- **The F1 Pilot's Struggle:** In a sharp turn, a pilot feels like their head is being ripped off toward the outside of the curve. In reality, their head simply wants to **continue in a straight line**, while the car (and the neck muscles) are forcing it to turn.

"Inertial forces are nothing more than the reflection of describing motion from a non-inertial, accelerated perspective."

How Did All Begun?

Recent genomic studies show that during a window of time around 43,500 and 50,500 years ago, Homo sapiens and Neanderthals mingled and interbred. We might be pretty sure that at such early times they looked at the night sky and wondered ...

Everyone Studied the Heavens

With time and careful observation many civilizations created their heavens models.

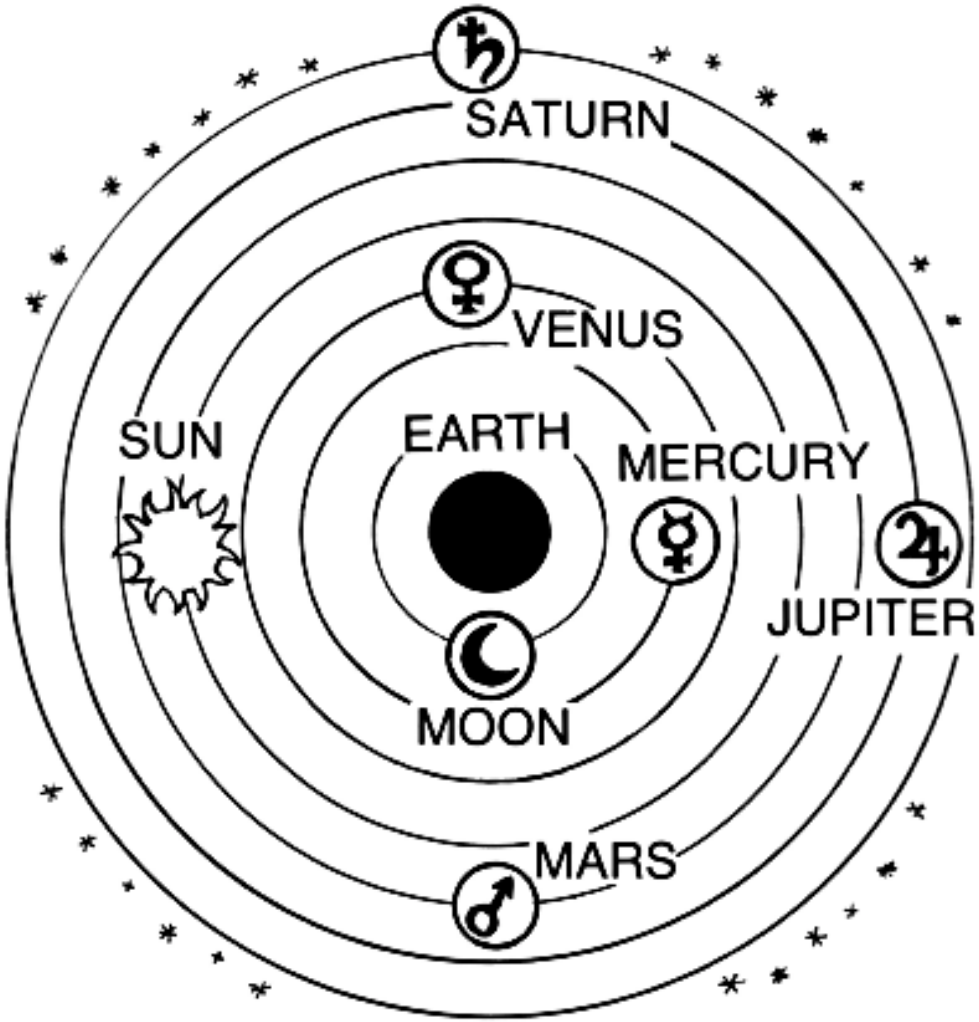
- A remarkable example, the **Dresden Codex**, a Pre-Columbian computer of Mayan origin, dated to the 11th or 12th century is a copy of a much older 5th-century text.
- The Dresden Codex an accordion-fold manuscript made of amate, is nothing less than an advanced astronomical almanac and "ephemeris."
- It contains high-precision tables for predicting solar eclipses and the 584-day synodic cycle of Venus with an error of only 0.08 days per century.
- The Codex proves that complex planetary dynamics were understood and mathematically modeled in the Americas long before the telescope.

Dresden Codex

The Geocentric Model

Claudius Ptolemy lived and worked in Alexandria, Egypt, during the **2nd century AD**. Ptolemy, is historically recognized as Greek—or at least an ethnically Hellenized Egyptian who wrote in Koine Greek and was part of the Greek intellectual community. Most historians agree that Ptolemy made extensive use of the Library of Alexandria and its vast resources to synthesize the centuries of scientific knowledge found in his major works, such as the Almagest and Geography.

Ptolemy's genius was not just in observation, but in mathematical synthesis. He transformed centuries of Hellenistic astronomy into a single, cohesive geometry: the Earth at rest, with the heavens moving in a complex dance of epicycles and deferents, circles within circles.


SATURN
VENUS
SUN
EARTH
MERCURY
JUPITER
MOON
MARS

Heliocentrism

Nicolaus Copernicus (**1473-1543**) was a Polish astronomer. He proposed the heliocentric theory, which stated that the Sun, not the Earth, was the stationary center of the universe, with the Earth revolving annually around the Sun and rotating daily on its axis.

Nearly 1,400 years for this change. During this time, the Earth was established as the immovable center. Both the Church and academia embraced it as absolute truth, for it not only 'explained' the heavens but also aligned perfectly with theology.

Ptolemy gave us a universe that made sense to our senses; Copernicus gave us a universe that made sense to our reason.

Kepler's Laws, A Case for Data Science

Johannes Kepler redefined the cosmos by discovering that planets move in ellipses—not circles—around the sun. He blended Tycho Brahe's unmatched data with bold mathematical insight. His three planetary laws grounded heliocentrism in observable precision and transformed astronomy into physical science.

Mathematics The Supreme Synthetizer.

Kepler's Laws

- Planetary orbits are elliptic shaped
- The radius vector form the sun to a planet sweeps equal areas in equal times
- The cube of the radius of the orbit is proportional to the square of the period

Kepler's Third Law: The Solar System Harmony ($1\,AU = 150 \times 10^6$ km)

Planet	Radius r (AU)	Period T (years)	Ratio r^3/T^2
Mercury	0.387	0.241	0.998
Venus	0.723	0.615	0.999
Earth	1.000	1.000	1.000
Mars	1.524	1.881	1.001
Jupiter	5.203	11.862	1.001
Saturn	9.537	29.457	1.000
Uranus	19.191	84.011	1.001
Neptune	30.069	164.79	1.001

Note: The ratio is nearly identical for all planets, proving Kepler's Third Law works across the entire Solar System!

Sidereus Nuncius
The Starry Messenger

SIDEREVS
NVNCIVS
MAGNA, LONGEQVE ADMIRABILIA
Spectacula pandens, suspiciendaque proponens
vnicuique, præsertim verò
PHILOSOPHIS, atq; ASTRONOMIS, quæ à
GALILEO GALILEO
PATRITIO FLORENTINO
Patauini Gymnasij Publico Mathematico
PERSPICILLI
Nuper à se reperti beneficio sunt obseruata in LVNÆ FACIE, PLANIS
NEBVLIS, LACTEO CIRCVLO, STELLIS NEBVLOSIS,
Appiritè verò in
QVATVOR PLANETIS
Circa IOVIS Stellam disparibus interuallis, atque periodis, celeri-
tate mirabili circumuolutis, quos, nemini in hoc tempo-
dum cognitos, nouissimè Author depre-
hendit primus, atque
MEDICEA SIDERA
NVNCVPANDOS DECREVIT.
VENETIIS, Apud Thomam Baglionum. M DC X.
Superiorum Permissu, & Priuilegio.

Galileo Galilei (1564-1642) telescopic observations in the early 17th century revolutionized astronomy by providing critical evidence for the heliocentric (Sun-centered) model. He discovered the four largest moons of Jupiter, the phases of Venus, craters on the Moon, sunspots, and that the Milky Way is made of stars.

Standing on the shoulders of giants, Newton (1642-1727) was able to improve our whole understanding of -not only the solar system but- the universe by postulating his famous *Univeral Law of Gravitation*

Newton's law of gravitation

Statement

Any particle of matter in the universe attracts any other with a force varying directly as the product of the masses and inversely as the square of the distance between them.

$$\boxed{\vec{F}_{12} = -G\frac{M_1 M_2}{r_{12}^2}\hat{\mathbf{r}}_{12}}$$

In the SI system

$$G = 6.674 \times 10^{-11}\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{kg}^2$$

MATH: A Simple Exercise

If we think of a planet in a circular orbit, the gravitational attraction of the sun will essentially be the only force acting on the planet and therefore [why?]

$$m\frac{v^2}{r} = m\frac{1}{r}\left(\frac{2\pi r}{T}\right)^2 = G\frac{Mm}{r^2}$$

This can be written as

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2},$$

since GM is constant for all solar system planets, then for any two planets

$$\frac{r_1^3}{T_1^2} = \frac{r_2^3}{T_2^2},$$

Which is Kepler's 3rd Law

The Lost Calculation: Newton and Halley

In **August 1684**, Edmond Halley visited **Cambridge** with a question that would change science forever.

- **The Query:** Halley asked what curve a planet would trace if gravity followed an inverse-square law.
- **The Surprise:** Newton immediately answered: **"An Ellipse."**
- **The Missing Proof:** Stunned, Halley asked for the calculation. Newton looked through his papers at his Cambridge rooms but *could not find it*.
- **The Legacy:** Newton promised to redo it. This spark led him to write the *Principia*, published thanks to Halley's persistence.

This encounter at Cambridge was the spark that led Newton to write the Principia.

Predictive Power: The Discovery of Neptune

By 1845, astronomers recognized seven planets. The farthest known was **Uranus**, discovered by **William Herschel** in **1781**.

- **The Anomaly:** Uranus wasn't following its predicted Newtonian path perfectly.
- **The Dilemma:** Was Newton's theory wrong, or was something else tugging at the planet?
- **The Hypothesis: John Couch Adams** and **Urbain Le Verrier** independently calculated that an unseen eighth planet must exist.
- **The Triumph:** Using Newton's mechanics, they predicted its position. In **1846**, Neptune was found exactly where they said it would be!

Newtonian mechanics didn't just explain the seen; it predicted the unseen.

Our Typical Sci Project

Object	**Orbital Radius / Physical Radius** [AU]
Sun (Radius)	0.00465
Mercury	0.39
Venus	0.72
Earth	1.00
Mars	1.52
Jupiter	5.20
Saturn	9.54
Uranus	19.22
Neptune	30.06

Table: Solar radius and average planetary distances in Astronomical Units (AU).

$1\,\text{AU} = 149,597,870,700 \text{ m} \approx 150 \text{ million km} = 92,955,807.3 \text{ miles}$

The Solar System at LSC-CyFair Scale

Object	Feet	Miles	Where would it be? (From HSC)
Sun (1 ft)	**1.00**	—	**HSC Building Entrance**
Mercury	83.87	0.016	Main Hallway of the HSC
Venus	154.84	0.029	Walking towards the Parking Lot
Earth	215.05	0.041	On the Bridge over the Lake
Mars	326.88	0.062	At the Library Entrance
Jupiter	1,118.28	0.21	On Barker Cypress Rd
Saturn	2,051.61	0.39	At the Berry Center
Uranus	4,133.33	0.78	At the Target on Hwy 290
Neptune	6,464.52	1.22	At the Towne Lake Boardwalk
Oort Cloud (End)	—	**4,073**	**In Lima, Peru!**

Table: Solar System Scale: If the Sun's radius were 1 foot.

"If the Sun is a 1-foot ball in our classroom, Neptune is at the Boardwalk!"

Going Beyond: The Milky Way Scale

- **Sun Radius:** 1 foot (The "beach ball" at HSC Building).
- **Distance to Neptune:** 1.22 miles (Towne Lake Boardwalk).
- **Edge of Oort Cloud:** 4,073 miles (Lima, Peru).
- **Milky Way Radius: 136,368,360 miles!**

The "Mind-Blowing" Fact:

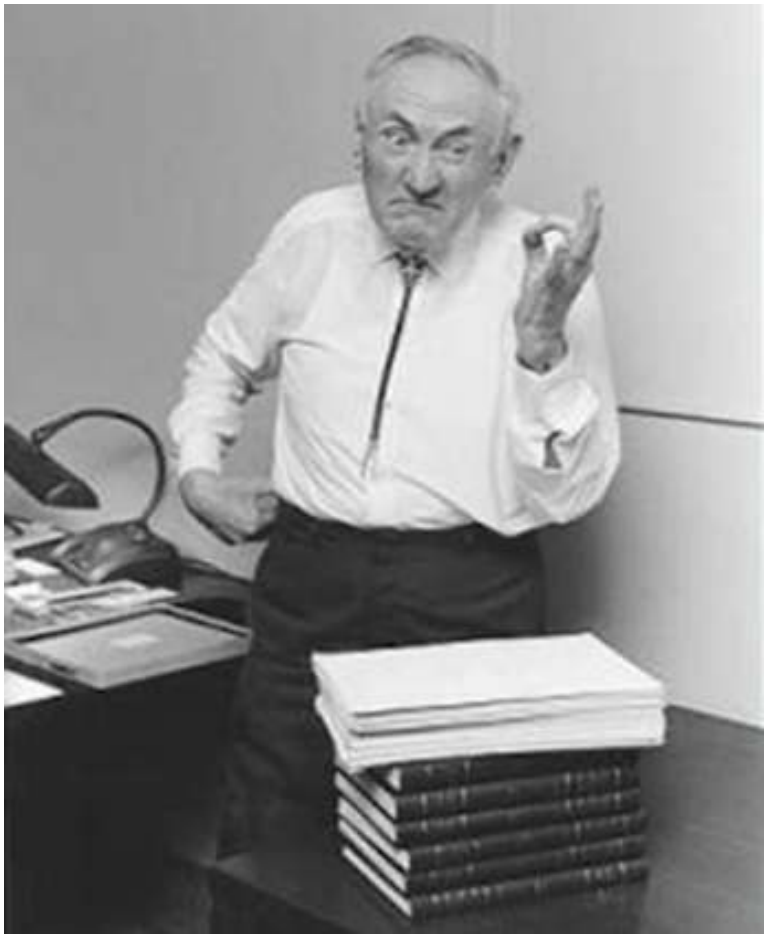
If our Sun is 1 foot wide at CyFair, the edge of our Galaxy is so far away that you would have to circle the entire Earth **5,477 times** to reach it!

Dunkle Materie: The Pain in The Neck of Astrophysics

Newton and Kepler gave us the rules, but in 1933, **Fritz Zwicky** a brilliant but "difficult" Swiss astronomer at Caltech.noticed the math didn't add up.

- Zwicky was infamous for his volatile temperament and his habitual disrespect for his contemporaries, frequently resorting to extremely crude and derogatory name-calling.
- **Studying** the **Coma Cluster** (a massive group of galaxies), he measured the galaxies' velocities and found they were moving too fast!
- **The Verdict:** According to Newton's Law of Gravity, the visible mass wasn't enough to hold them together. They should have flown apart.

Zwicky and Rubin, the Dynamic Duo 40 years appart



Zwicky and Rubin, the Dynamic Duo

Dunkle Materie

Zwicky concluded that 90% of the mass was invisible. He coined the term **Dark Matter**.
Nobody believed him for 40 years... until

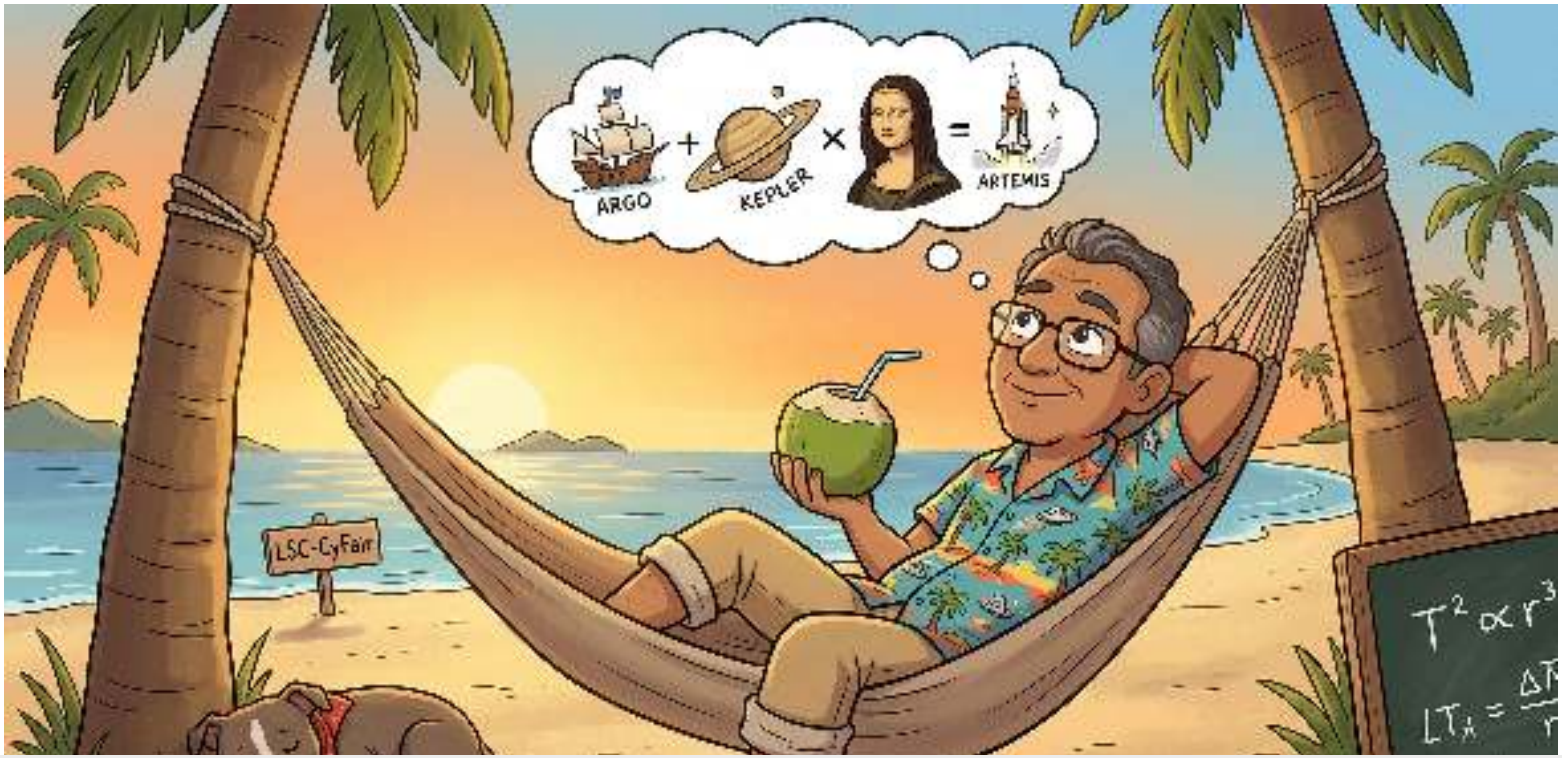
In 1970, **Vera Rubin**, from the Carnegie Institution ws studying the motion of stars in galaxies (in Andromeda in particular) and found that the rotation velocity of the stars within a galaxy was different than expected showing that Newton was only able to explain 10% of the story which strongly supported Zwicky proposal.

The Aftermath: Dark Matter and Dark Energy

Our current understanding suggests that **Dark Matter** is essential for our gravitational theories to remain consistent with observations.

- **Dark Matter:** Named because it does *not* emit, absorb, or reflect electromagnetic radiation; otherwise, we could detect it with our telescopes.
- **The Evidence:** Further extragalactic and cosmological studies not only support Dark Matter but also point to an even more exotic phenomenon: **Dark Energy**.
- **Dark Energy:** Required to explain the observed fact that the **expansion of the universe is accelerating**.

We are currently staring at a universe where 95% *of its content remains a mystery (our ignorance is staggering to say the least).*

A Simple Thought

The same math that Newton used to explain an apple falling, and that Kepler used to map the ellipses of the giants, is what is taking us back to the stars today. Gravity is not a law we obey; it's the map we use to navigate the infinite.